=== UNIT 1: ELECTROMAGNETIC THEORY & MAXWELL'S EQUATIONS ===

>> 1.1 Vector Calculus Foundations for Electromagnetism

Classical electromagnetic theory provides the unified physical description of electricity, magnetism, and light. In civil engineering and surveying, EM theory underpins radar subsurface imaging, GPS geodesy, and structural non-destructive sensing.

* Scalar and Vector Fields: A scalar field assigns a single numeric magnitude phi(x,y,z) to every spatial point (e.g., electric potential V, temperature T). A vector field assigns magnitude and spatial direction F(x,y,z) (e.g., electric field E, magnetic flux density B).
* Del (Nabla) Operator: The differential spatial operator defined as: Del = i * (d/dx) + j * (d/dy) + k * (d/dz).
* Gradient of a Scalar Field (grad phi = Del phi): A vector pointing in the direction of maximum spatial increase of phi, with magnitude equal to the rate of increase. In electrostatics: E = - Del V.
* Divergence of a Vector Field (div F = Del dot F): A scalar measure of net outwards flux per unit volume from an infinitesimal region around a point. div F > 0 signifies a source; div F < 0 denotes a sink; div F = 0 indicates a solenoidal field.
* Curl of a Vector Field (curl F = Del x F): A vector measuring the microscopic circulation or vortex density of field lines around a point. If Del x F = 0, the field is irrotational (conservative).
* Gauss's Divergence Theorem: Transforms a volume integral of divergence into a closed surface integral: Integral_V (Del dot F) dV = Closed_Integral_S (F dot dS).
* Stokes's Theorem: Transforms an open surface integral of curl into a closed line integral: Integral_S ((Del x F) dot dS) = Closed_Integral_C (F dot dr).

>> 1.2 Equation of Continuity and Inconsistency of Ampere's Law

* Principle of Charge Conservation: Electric charge can neither be created nor destroyed. The net outward current leaving a closed volume equals the rate of decrease of enclosed charge:

Equation of Continuity: Del dot J = - d(rho) / dt, where J is current density (A/m^2) and rho is volume charge density (C/m^3).

* Inconsistency of Ampere's Circuital Law: Ampere's original static law states: Del x B = mu_0 * J. Taking the divergence of both sides: Del dot (Del x B) = mu_0 * (Del dot J).

By vector calculus identity, the divergence of any curl is identically zero: Del dot (Del x B) = 0. This requires Del dot J = 0, which holds strictly for steady DC currents where d(rho)/dt = 0, but fails completely for time-varying (AC) electromagnetic fields where charge density changes dynamically.

>> 1.3 Maxwell’s Displacement Current and the Complete Maxwell Equations

To resolve the inconsistency, James Clerk Maxwell postulated that a time-varying electric field generates an equivalent magnetic effect termed Displacement Current Density J_d:
* Formulation: From Gauss’s Law Del dot D = rho, differentiating with respect to time: d(rho)/dt = Del dot (dD/dt). Substituting into continuity equation: Del dot J + Del dot (dD/dt) = 0 => Del dot (J + dD/dt) = 0.
* Displacement Current Density: J_d = dD/dt = epsilon_0 * (dE/dt). Unlike conduction current J (conveyed by physical drift of charge carriers), displacement current produces magnetic fields through dynamic variation of electric displacement D across dielectrics or vacuum.
* Maxwell’s Equations in Differential Form (Free Space / Media):
- (1) Gauss’s Law for Electrostatics: Del dot D = rho_free (or Del dot E = rho / epsilon_0).
- (2) Gauss’s Law for Magnetism: Del dot B = 0 (No isolated magnetic monopoles exist).
- (3) Faraday’s Law of Induction: Del x E = - dB / dt (Time-varying magnetic flux induces electric circulation).
- (4) Ampere-Maxwell Circuital Law: Del x H = J_free + dD / dt (Conduction and displacement currents produce magnetic curl).

>> 1.4 Electromagnetic Wave Propagation in Dielectric Media

In a source-free, charge-free dielectric medium (rho = 0, J = 0):
* Wave Equations: Taking curl of Faraday’s law: Del x (Del x E) = - d/dt (Del x B).
Using vector identity Del x (Del x E) = Del(Del dot E) - Del^2(E):
Since Del dot E = 0: - Del^2(E) = - mu * epsilon * (d^2E / dt^2) => Del^2(E) = mu * epsilon * (d^2E / dt^2).
Similarly for magnetic field: Del^2(H) = mu * epsilon * (d^2H / dt^2).
* Phase Velocity of EM Waves: v = 1 / sqrt(mu * epsilon). In free vacuum: c = 1 / sqrt(mu_0 * epsilon_0) = 2.99792 * 10^8 m/s.
* Transverse Nature of EM Waves: Plane EM waves travelling along z-direction have E_z = 0 and H_z = 0. Electric field E, magnetic field H, and wave propagation vector k are mutually orthogonal: E perpendicular to H perpendicular to k.
* Characteristic Wave Impedance: Z_0 = |E| / |H| = sqrt(mu_0 / epsilon_0) = sqrt(4*pi*10^-7 / 8.854*10^-12) = 376.73 Ohms (~ 377 Ohms).

>> 1.5 Poynting Vector and Poynting’s Theorem

* Poynting Vector S: Represents the directional rate of electromagnetic energy transfer per unit cross-sectional area (Watts/m^2):
S = E x H = (1 / mu) * (E x B).
* Physical Direction: The energy flows in the direction of vector cross-product E x H, coinciding with the wave propagation vector k.

* Time-Average Poynting Vector (Irradiance / Intensity): S_avg = 0.5 * Re(E x H*) = 0.5 * (E_0^2 / Z_0) = 0.5 * epsilon_0 * c * E_0^2.
* Poynting's Theorem: Mathematical statement of energy conservation for electrodynamics: - Closed_Integral (S dot dS) = d/dt Integral_V [0.5*epsilon*E^2 + 0.5*mu*H^2] dV + Integral_V (J dot E) dV. (Net inward energy flux equals rate of increase of stored EM energy plus ohmic Joule dissipation).

>> 1.6 EM Waves in Conducting Media and Skin Depth

In a lossy conducting medium with conductivity sigma:
* Propagation Constant: gamma = alpha + j*beta = sqrt(j * omega * mu * (sigma + j*omega*epsilon)).
* Good Conductor Approximation (sigma >> omega*epsilon): alpha ~= beta ~= sqrt(pi * f * mu * sigma).
* Skin Depth (delta): The penetration depth at which EM wave amplitude decays to 1/e (~36.8%) of its surface boundary value:
Formula: delta = 1 / alpha = sqrt(2 / (omega * mu * sigma)) = 1 / sqrt(pi * f * mu * sigma).

[EXAMPLE 1: Displacement Current in Charging Circular Capacitor]

Problem Context: A parallel-plate circular capacitor has plates of radius R = 100 mm separated by distance d = 2.0 mm in air (epsilon_0 = 8.854 * 10^-12 F/m). It is charged by an AC voltage V(t) = 230 * sqrt(2) * sin(100 * pi * t) Volts (50 Hz). Calculate: (1) Capacitance C, (2) Conduction current i_c entering the plates, (3) Displacement current density J_d between the plates, (4) Magnetic field B at radius r = 50 mm from central axis.
Solution Step 1: Capacitance C:
Area A = pi * R^2 = pi * (0.10)^2 = 0.031416 m^2.
C = (epsilon_0 * A) / d = (8.854 * 10^-12 * 0.031416) / 0.002 = 1.3908 * 10^-10 F = 139.08 pF.
Solution Step 2: Conduction Current i_c(t):
i_c(t) = C * (dV/dt) = C * [230 * sqrt(2) * (100 * pi) * cos(100 * pi * t)].
Peak conduction current: I_c_peak = 1.3908 * 10^-10 * 325.27 * 314.16 = 1.421 * 10^-5 A = 14.21 microamps.
Solution Step 3: Displacement Current Density J_d:
Electric field between plates: E(t) = V(t) / d.
J_d = epsilon_0 * (dE/dt) = (epsilon_0 / d) * (dV/dt) = (C / A) * (dV/dt) = i_c(t) / A.
Peak J_d = I_c_peak / A = 1.421 * 10^-5 A / 0.031416 m^2 = 4.523 * 10^-4 A/m^2 (452.3 microamps/m^2).
Total displacement current i_d = J_d * A = I_c_peak = 14.21 microamps (Displacement current matches conduction current identically!).
Solution Step 4: Magnetic field B at radius r = 50 mm (r < R):

Using Ampere-Maxwell law along circular path of radius r:
Closed_Integral B dot dl = mu_0 * i_d_enclosed => B * (2 * pi * r) = mu_0 * [J_d * (pi * r^2)].
B = (mu_0 * J_d * r) / 2 = (4 * pi * 10^-7 * 4.523 * 10^-4 * 0.050) / 2 = 1.421 * 10^-11 Tesla.
Conclusion: Peak displacement current = 14.21 microamps; Magnetic field at r=50mm is B = 1.421 * 10^-11 T.

[EXAMPLE 2: Verification of Maxwell's Second Law for Magnetic Field]

Problem Context: A magnetic field in an electrical transformer core is modeled as B(x,y,z) = (2*x*y) i + (y^2 - x^2) j + (C*y*z) k. Using Maxwell's equations, determine constant C such that B constitutes a physically admissible magnetic field.
Solution Step 1: Maxwell's Second Law Condition:
Gauss's Law for magnetism states that magnetic fields are strictly solenoidal: div B = Del dot B = 0.
Solution Step 2: Compute Divergence of B:
Del dot B = d(Bx)/dx + d(By)/dy + d(Bz)/dz.
d(Bx)/dx = d(2*x*y)/dx = 2*y.
d(By)/dy = d(y^2 - x^2)/dy = 2*y.
d(Bz)/dz = d(C*y*z)/dz = C*y.
Del dot B = 2*y + 2*y + C*y = (4 + C) * y.
Solution Step 3: Solve for C:
For Del dot B = 0 everywhere in space: (4 + C) * y = 0 => 4 + C = 0 => C = - 4.
Conclusion: The constant must be C = -4 to ensure no isolated magnetic monopoles exist.

[EXAMPLE 3: Solar Radiation Poynting Vector & Field Amplitudes]

Problem Context: Solar irradiance reaching the top of Earth's atmosphere is measured at S_avg = 1361 W/m^2 (the Solar Constant). Assuming the solar radiation can be approximated as a plane monochromatic EM wave in vacuum, calculate: (1) Peak electric field amplitude E_0, (2) Peak magnetic field amplitude H_0 and B_0, (3) Radiation pressure exerted on a perfectly reflecting civil solar collector panel.
Solution Step 1: Calculate Peak Electric Field E_0:
Average Poynting vector: S_avg = 0.5 * (E_0^2 / Z_0), where Z_0 = 376.73 Ohms.
E_0^2 = 2 * Z_0 * S_avg = 2 * 376.73 * 1361 = 1,025,459 (V/m)^2.
E_0 = sqrt(1,025,459) = 1012.65 V/m (approx 1.01 kV/m).
Solution Step 2: Calculate Peak Magnetic Fields H_0 and B_0:
H_0 = E_0 / Z_0 = 1012.65 / 376.73 = 2.688 A/m.
B_0 = mu_0 * H_0 = (4 * pi * 10^-7) * 2.688 = 3.378 * 10^-6 Tesla = 3.38 microTesla.
Solution Step 3: Calculate Radiation Pressure on Perfectly Reflecting Panel:
For total reflection, momentum transferred is doubled: P_rad = 2 * S_avg / c.

P_rad = (2 * 1361 W/m^2) / (3.0 * 10^8 m/s) = 2722 / 3.0 * 10^8 = 9.073 * 10^-6 N/m^2 = 9.07 microPascals.
Conclusion: Peak E_0 = 1012.65 V/m; Peak B_0 = 3.38 microTesla; Radiation pressure = 9.07 microPa.

[EXAMPLE 4: Skin Depth and Attenuation in Seawater for Subsurface Radar]

Problem Context: In marine civil engineering, ground-penetrating radar (GPR) operates at frequency f = 25 MHz in seawater (relative permittivity epsilon_r = 81, permeability mu = mu_0 = 4*pi*10^-7 H/m, conductivity sigma = 4.0 S/m). Calculate: (1) Conduction to displacement current ratio sigma / (omega * epsilon), (2) Skin depth delta, (3) Distance at which wave power drops by 40 dB.
Solution Step 1: Conduction to Displacement Current Ratio:
Angular frequency: omega = 2 * pi * f = 2 * pi * 25 * 10^6 = 1.5708 * 10^8 rad/s.
epsilon = epsilon_r * epsilon_0 = 81 * 8.854 * 10^-12 = 7.1717 * 10^-10 F/m.
omega * epsilon = 1.5708 * 10^8 * 7.1717 * 10^-10 = 0.1126 S/m.
Ratio = sigma / (omega * epsilon) = 4.0 / 0.1126 = 35.51 >> 1 => Behaves as a good conductor.
Solution Step 2: Compute Skin Depth delta:
delta = 1 / sqrt(pi * f * mu_0 * sigma) = 1 / sqrt(pi * 25*10^6 * 4*pi*10^-7 * 4.0) = 1 / sqrt(394.78) = 1 / 19.869 = 0.0503 m = 50.3 mm.
Solution Step 3: Distance for 40 dB Power Attenuation:
Attenuation in dB = 8.686 * alpha * x, where alpha = 1 / delta = 19.869 Np/m.
Attenuation rate = 8.686 * 19.869 = 172.58 dB/m.
For 40 dB drop: x = 40 / 172.58 = 0.2318 m = 23.2 cm.
Conclusion: Skin depth = 50.3 mm; EM power attenuates by 40 dB within just 23.2 cm, explaining why radio communication cannot penetrate deep ocean waters.

>> 1.7 University Review & Examination Questions

* Q1: Derive Maxwell’s four equations in differential and integral forms and explain the physical significance of each.
Answer: (1) Del dot D = rho: Gauss’s Law states that electric field flux diverging from any closed volume is proportional to the total enclosed free electric charge. Lines of E originate on positive and terminate on negative charges. (2) Del dot B = 0: Gauss’s Law for magnetism states that magnetic flux through any closed Gaussian surface is identically zero; magnetic field lines are continuous closed loops with no isolated magnetic poles. (3) Del x E = - dB/dt: Faraday’s Law states that a changing magnetic field creates a circulating electric field with non-zero curl, generating induced electromotive force. (4) Del x H = J + dD/dt: Ampere-Maxwell Law states that magnetic circulation is generated by both physical electric conduction currents and time-varying electric displacement currents.
* Q2: Explain Maxwell’s concept of displacement current and prove that displacement

current equals conduction current inside a charging capacitor.
Answer: Maxwell recognized that in AC circuits with capacitors, physical charge cannot jump across the dielectric gap, yet continuous current flows in external wires. To preserve charge conservation (Del dot J = -d(rho)/dt), Maxwell introduced displacement current J_d = dD/dt = epsilon * dE/dt. In a parallel plate capacitor of area A and charge q, electric field is E = q / (epsilon * A). The electric displacement is D = epsilon * E = q / A. Differentiating with respect to time: dD/dt = (1/A) * (dq/dt) = i_c / A, where i_c is conduction current in the leads. Total displacement current is: i_d = Integral (J_d dot dA) = J_d * A = (i_c / A) * A = i_c. Hence, displacement current between the plates exactly equals conduction current in the lead wires.

>> 1.8 Key Takeaways and Revision Points

* Displacement current J_d = dD/dt resolves the mathematical breakdown of Ampere’s law for AC fields.
* EM waves travel at speed v = 1/sqrt(mu*epsilon); in free vacuum c = 3 * 10^8 m/s.
* Characteristic impedance of free space is Z_0 = 377 Ohms.
* Poynting vector S = E x H quantifies directional electromagnetic energy flux per unit area.